

Lab - Deploying Containers to Azure Web Apps

To Begin with the Lab

Summary

This lab demonstrates how to deploy a Docker container stored in Azure Container Registry (ACR) to Azure Web Apps. Instead of publishing application source code, Azure Web Apps can directly host a Linux container image from ACR.

Understanding

Azure Web Apps supports two deployment methods:

- **Code Deployment** (ASP.NET, Node.js, Java, Python, PHP, etc.)
- **Container Deployment** (Docker images)

In this lab, the Docker image is stored in **Azure Container Registry (ACR)**. During Web App creation, the publish option is set to **Container**, and Azure automatically pulls the image from ACR and runs it.

Azure Web Apps automatically exposes **Port 80** to users. If your containerized application listens on **Port 8080**, Azure automatically forwards requests from **Port 80** to **Port 8080** inside the container, so no manual port mapping is required.

Example:

- User opens:

<https://mydockerapp.azurewebsites.net>

- Azure Web App receives traffic on:

Port 80

- Azure forwards the request to:

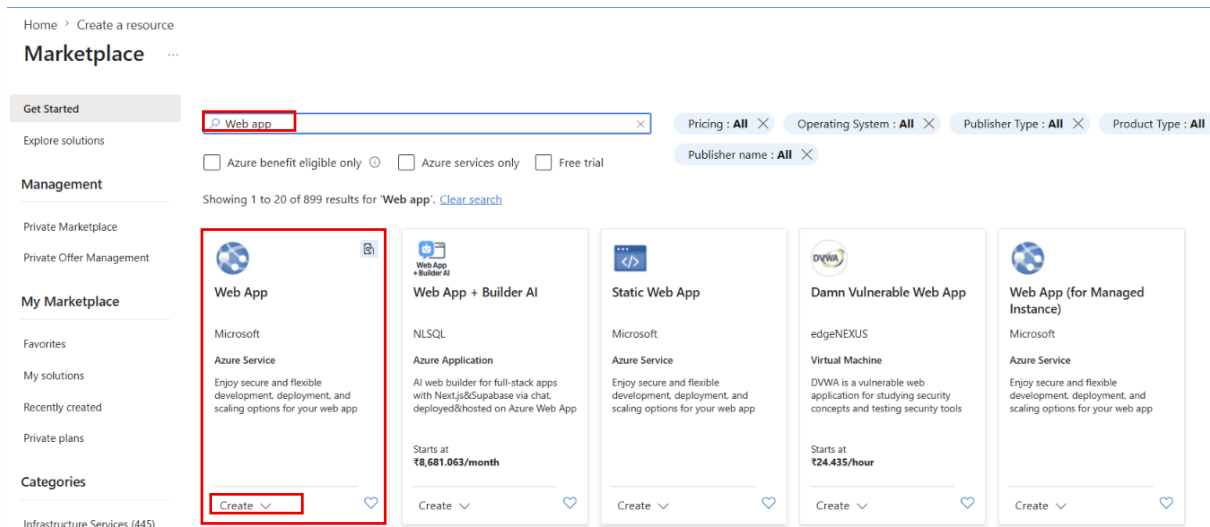
Container Port 8080

- The application running inside the container responds to the request.
-

Lab Steps

Step 1: Create an Azure Web App

- Open the Azure Portal.
- Search for **Web App**.
- Click **Create**.



- Select the existing **Resource Group**.
- Enter a unique **Web App Name**.
- Under **Publish**, select **Container**.
- Choose **Linux** as the operating system.
- Select the desired **Region** (East US in this lab).
- Choose the **Basic (B1)** App Service Plan.
- Click **Next**.

Home > Create a resource > Marketplace

Create Web App

on your resources

Subscription * Visual Studio Enterprise Subscription

Resource Group * Webapp [Create new](#)

Instance Details

Name docker-webapp ✓
-evh2euc0b0bnaqd2.eastus-01.azurewebsites.net

Publish * ☐ Code ☒ Container

Operating System * ☒ Linux ☐ Windows

Region * East US

Pricing plans
App Service plan pricing tier determines the location, features, cost and compute resources associated with your app.
[Learn more](#)

Linux Plan (East US) * (New) ASP-Webapp-931b [Create new](#)

Pricing plan Basic B1 (1.75 GB memory, 1 vCPU) [Explore pricing plans](#)

Step 2: Configure the Container

- Disable **Sidecar Support**.
- Select **Azure Container Registry (ACR)**.
- Enable **Admin Credentials**.

- Select the **Registry Domain Name**.
- Choose the **Docker App Image**.
- Select the required **Image Tag**.
- Leave **Startup Command** blank.
- Click **Next**.

Home > Create a resource > Marketplace

Create Web App ...

Basics Database **Container** Networking Monitor + secure Tags Review + create

Select your preferred source for container images. You can change these settings and other dependencies after creating the app. [Learn more](#)

Sidecar support ☒ Enhanced configuration with sidecar support off [Learn More](#)

Image Source * ☐ Quickstart ☒ Azure Container Registry ☐ Other container registries

Azure container registry options

Registry * regdevs12

Authentication * ☐ Managed identity ☒ Admin credentials

Image * dockerapp

Tag * v1

Startup Command ⓘ Example: /bin/bash; -c; echo hello; sleep 10000

Review + create

< Previous

Next : Networking >

Step 3: Configure Networking

- Keep the default **Modern Secure Networking** settings.
- Continue without making changes.

Home > Create a resource > Marketplace

Create Web App ...

Basics Database Container **Networking** Monitor + secure Tags Review + create

Web Apps can be provisioned with the inbound address being public to the internet or isolated to an Azure virtual network. Web Apps can also be provisioned with outbound traffic able to reach endpoints in a virtual network, be governed by network security groups or affected by virtual network routes. By default, your app is open to the internet and cannot reach into a virtual network. These aspects can also be changed after the app is provisioned. [Learn more](#)

Enable public access * ⓘ ☒ On ☐ Off

Enable virtual network integration * ⓘ ☐ On ☒ Off

Step 4: Skip Application Insights

- Do not enable **Application Insights**.
- Click **Review + Create**.

Create Web App ...

Basics Database Container Networking Monitor + secure Tags **Review + create**

Summary



Web App
by Microsoft

Basic (B1) sku

Estimated price - 1032.45 INR/Month



Basic authentication for this app is currently disabled and may impact deployments. Click to learn more.

Details

Subscription	294539c9-c109-4db8-b785-0043716c0b11
Resource Group	Webapp
Name	docker-webapp
Secure unique default hostname	Enabled
Publish	Container
Image:Tag	regdeves12.azurecr.io/dockerapp:v1
Server URL	https://regdeves12.azurecr.io

App Service Plan (New)

Name	ASP-Webapp-931b
Operating System	Linux

Create

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Step 5: Create the Web App

- Review all configuration settings.
- Click **Create**.

Create Web App ...

Basics Database Container Networking Monitor + secure Tags Review + create

Summary



Web App
by Microsoft

Basic (B1) sku

Estimated price - 1032.45 INR/Month



Basic authentication for this app is currently disabled and may impact deployments. Click to learn more.

Details

Subscription	294539c9-c109-4db8-b785-0043716c0b11
Resource Group	Webapp
Name	docker-webapp
Secure unique default hostname	Enabled
Publish	Container
Image:Tag	regdeves12.azurecr.io/dockerapp:v1
Server URL	https://regdeves12.azurecr.io

App Service Plan (New)

Name	ASP-Webapp-931b
Operating System	Linux

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- Wait until deployment completes successfully.

Home



Microsoft.Web-WebApp-Portal-9b483247-b638 | Overview

Deployment

Search Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name: Microsoft.Web-WebApp-Portal-9b483247-b638
Subscription: Visual Studio Enterprise Subscription
Resource group: Webapp

Start time: 7/23/2026, 2:06:35 PM

Correlation ID: bd23c220-e17b-4224-905f-1d28dff2ac6d

Deployment details

Next steps

[Manage deployments for your app.](#) Recommended

[Protect your app with authentication.](#) Recommended

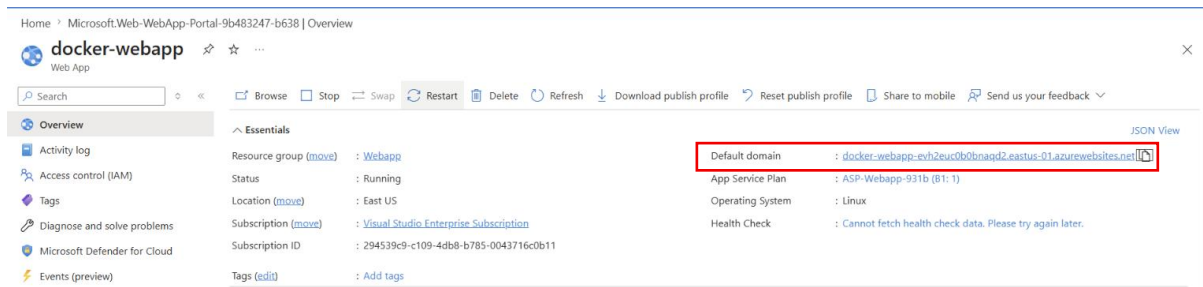
[Go to resource](#)

Give feedback

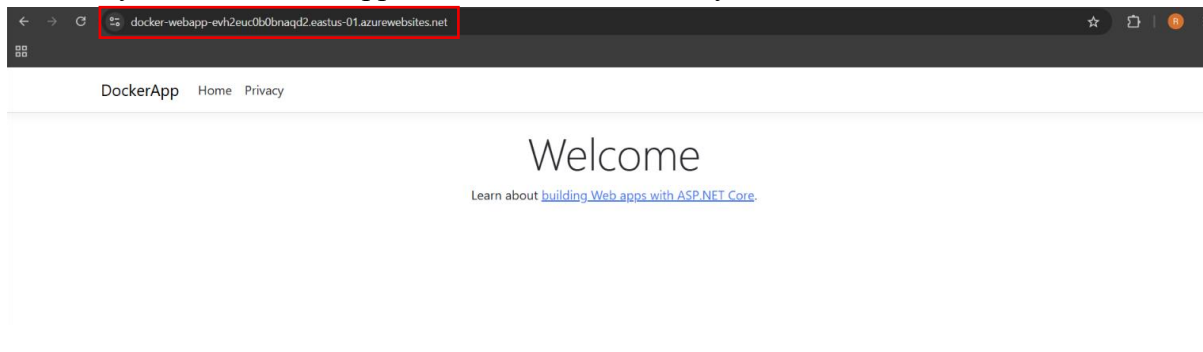
[Tell us about your experience with deployment](#)

Step 6: Access the Application

- Open the newly created **Web App**.
- Copy the **Default Domain Name**.



- Open it in a web browser.
- Wait a minute while the container starts.
- Verify that the Docker application loads successfully.



Step 7: Understand Port Mapping

- Azure Web Apps receives requests on **Port 80**.
- If the container listens on **Port 8080**, Azure automatically forwards traffic.
- No Docker -p command or manual port mapping is required.